

Why Custom Agents?

You now have rules (steering) and procedures (skills). But right now, Kiro's default agent handles everything. What if you want a dedicated agent identity specifically for EOL reporting, with its own system prompt and configuration? That's what custom agents are for. In Section 3, you'll see advanced agent patterns like `includePowers` and `parallel` subagent invocation.

Custom agents provide a way to customize Kiro behavior by defining specific configurations for different use cases. Each custom agent is defined by a configuration file that specifies which tools the agent can access, what permissions it has, and what context it should include.

Learn more: [CLI Custom Agents docs](#) 📖

You now have rules (steering) and procedures (skills). But right now, Kiro's default agent handles everything. What if you want a dedicated agent identity specifically for EOL reporting, with its own system prompt and configuration? That's what custom agents are for.

Custom agents allow you to:

- **Pre-approve specific tools** — Define which tools can run without prompting
- **Limit tool access** — Restrict which tools are available to reduce complexity
- **Include relevant context** — Automatically load project files, documentation, or system information
- **Configure tool behavior** — Set specific parameters for how tools should operate

When to Use Each Layer

Scenario	Use
General rules for all conversations	Steering
Reusable procedure for a specific task	Skill
Dedicated identity with specific behavior	Agent
Different agents for different domains	Agent per domain

Step 3.1: Create the Agent File

Custom agents in Kiro are markdown files stored in `.kiro/agents/` (workspace) or `~/kiro/agents/` (global). Each agent has:

- A **description** in YAML frontmatter — tells Kiro when to auto-select this agent
- A **system prompt** in the markdown body — defines the agent's identity, behavior, and instructions

Create the agents directory:

```
1 mkdir -p .kiro/agents
```

IDE (.md)CLI (.json)

Create .kiro/agents/eol-reporting.md:

```
1 cat > .kiro/agents/eol-reporting.md << 'EOF'
2 ---
3 name: eol-reporting
4 description: >
5   AI agent specialized in tracking End of Service (EOS) and End of Life (EOL) dates
6   for AWS cloud infrastructure resources. Use when asked to generate EOL reports,
7   check deprecation dates, inventory AWS resource versions, or assess lifecycle risks.
8 tools: ["read", "write", "shell", "web"]
9 model: auto
10 includeMcpJson: >
11   {
12     "aws-knowledge-mcp": {
13       "url": "https://knowledge-mcp.global.api.aws",
14       "disabled": false,
15       "autoApprove": ["aws__search_documentation", "aws__read_documentation"]
16     }
17   }
18 ---
19
20 # EOS/EOL Reporting Agent
21
22 You are an AI agent specialized in tracking EOS/EOL dates for cloud infrastructure resources.
23
24 ## Core Workflow
25 1. Inventory AWS Resources – Use AWS CLI
26 2. Retrieve EOL Data – Use the `retrieve-eol-data` skill
27 3. Categorize by Urgency – EXPIRED / Critical / Warning / Monitor
28 4. Generate Reports – Use branded templates
29
30 ## Rules
31 - AWS official documentation is always the primary source
32 - Format dates consistently (YYYY-MM-DD)
33 - Include resource identifiers (ARN, name, region)
34
35 ## Data Source Priority
36 1. AWS Official Documentation (via Knowledge MCP)
37 2. AWS Health Dashboard
38 3. AWS CLI APIs
39 4. endoflife.date API (non-AWS only)
40
41 ## Report Templates
42 - Markdown: #[[file:.kiro/agents/references/eol-report/eol-report-template.md]]
43 - HTML: #[[file:.kiro/agents/references/eol-report/eol-report-template.html]]
44 EOF
```

🔑 Key Points

- The description in YAML frontmatter is what Kiro uses for auto-selection — include trigger phrases like "EOL report", "deprecation dates"
- The markdown body is the system prompt — it defines the agent's identity and full instructions
- `includeMcpJson` embeds the MCP server config directly in the agent file

Step 3.2: Refactor the Steering File

Now that the agent owns the full rulebook, the steering file should slim down to just a routing pointer:

```
1 cat > .kiro/steering/eol-reporting.md << 'EOF'
2 ---
3 inclusion: always
4 ---
5 # EOS/EOL Reporting Agent
6
7 For any End of Service (EOS) and End of Life (EOL) topics – including generating EOL reports,
8 checking deprecation dates, inventorying AWS resource versions, or assessing lifecycle risks –
9 you must use the `eol-reporting` custom agent.
10 EOF
```

Why keep the steering file? It's always loaded into every conversation. By keeping a slim pointer, any conversation will know to route EOL questions to the right agent.

Step 3.3: Move Report Templates

Since the agent now owns the report configuration, move templates to the agent's references folder:

```
1 mkdir -p .kiro/agents/references/eol-report
2 mv .kiro/steering/eol-report/* .kiro/agents/references/eol-report/
3 rmdir .kiro/steering/eol-report
```

Step 3.4: Test the Agent

Before testing, start a fresh session so Kiro picks up the new agent.

IDE (.md)CLI (.json)

Kiro should auto-select the `eol-reporting` agent when your request matches its description:

```
1 Generate a weekly EOL report for my AWS infrastructure
```

You can also invoke it explicitly using the agent selector in the chat panel.

Step 3.5: Verify the Complete Architecture

Your final directory structure should look like this:

```
.kiro/
├── agents/
│   ├── eol-reporting.md          # Agent identity & prompt (IDE)
│   ├── eol-reporting.json       # Agent config for CLI
│   └── references/
│       ├── eol-report/
│       │   ├── eol-report-template.html
│       │   ├── eol-report-template.md
│       │   └── logo-2c2p-d.webp
│       └──
├── skills/
│   ├── retrieve-eol-data/
│   │   ├── SKILL.md
│   │   ├── references/
│   │   └── data-sources.md
│   └──
├── steering/
│   └── eol-reporting.md         # Agent routing pointer
```

Test the complete flow:

```
1 Generate a weekly EOS/EOL report for my AWS infrastructure. Include:
2 1. Summary section with counts by category
3 2. Critical resources section (< 6 months)
4 3. Warning resources section (6-12 months)
5 4. Monitor resources section (1-2 years)
6 5. Recommendations section with prioritized actions
```

The agent owns the rules and report format. The steering routes EOL questions to the agent. The skill provides the procedures. All three layers work together seamlessly.

Final Architecture



Final Directory Structure

```
.kiro/
├── agents/
│   ├── eol-reporting.md          # Agent identity (IDE)
│   ├── eol-reporting.json       # Agent config (CLI)
│   └── references/
│       ├── eol-report/
│       │   ├── eol-report-template.html
│       │   ├── eol-report-template.md
│       │   └── logo-2c2p-d.webp
│       └──
├── skills/
│   ├── retrieve-eol-data/       # Reusable skill
│   │   ├── SKILL.md
│   │   ├── references/
│   │   └── data-sources.md
│   └──
├── steering/
│   └── eol-reporting.md         # Agent routing pointer
```

The Story Arc

1. **Steering** — "I taught Kiro the rules" → Everything works, but the file is bloated with both rules and procedures
2. **Skills** — "I packaged the procedures so they load only when needed" → Context is lean, procedures are reusable
3. **Custom Agent** — "I built a dedicated agent that owns the rules and ties it all together" → A purpose-built identity with full rulebook, steering becomes a routing pointer

Troubleshooting

Issue	Solution
Agent can't access AWS Knowledge MCP	Verify <code>~/kiro/settings/mcp.json</code> configuration
Skill doesn't activate automatically	Check <code>description</code> field includes trigger phrases; start a fresh session
Agent isn't auto-selected	Check <code>description</code> in YAML frontmatter includes relevant keywords
EOL dates are incorrect	Verify AWS official documentation is being used as primary source
AWS Health API returns <code>SubscriptionRequiredException</code>	Account lacks Business/Enterprise Support — agent should continue with other sources

🔑 What's Next

You've built the complete 3-layer stack: Steering -> Skill -> Agent. In Section 3, you'll add a 4th layer (Powers) and learn advanced patterns for each layer using a Terraform infrastructure review use case.